MONTHLY PROGRESS REPORT

Month of: January, 2026

PGSF/UGSF - Fellowship scheme for student of Devang Patel Institute of Advance Technology and Research (Information Technology)

1) Name of the Fellow: Kartik Guleria
2) Date of commencement of work: 01/01/2026
3) Name of the Institute/Department/Center/Cell under which he/she is working: Information Technology
4) Total number of working hours during the period: 24 hours
5) Number of days the Fellow remained present at the University/Institution 23 days
6) Details of the work done during the period: **Annexure A**
7) Total Fellowship Amount claimed Rs. **Rupees Four Thousand Six Hundred Only** (Rupees in word)
8) Bank details of Fellow:

Name of Bank : Central Bank of India

Account No. : 3683449377

Address of Branch : GIDC ESTATE , OLD NATIONAL HIGHWAY NO 8 , ANKLESHWAR EAST - 393002

IFSC Code : CBIN0282558

Comments of the supervisor on the progress of the work assigned during the period under report: The work was completed on time with good accuracy and dedication.

Date: _____

Fellow Name : Kartik Guleria

Fellow Signature:

Fellow Std. ID No: 23DIT015

Name of the Supervisor : Ashish Katira **Signature :**

Signature (Principal/HoD)	Signature (Dy. Registrar)	Signature (Dean Research)

Annexure A (Details of Work done)

Sr No	Date	Work Done	Hours	Sign of Student
1	WEEK 1	• Developed student export API (/api/students/export) with GET/POST endpoints supporting department, semester, and admission year filtering and cross-collection join with ProjectGroup for project and guide metadata in the exported .xlsx file. • Implemented guide export API (/api/guides/export) with dynamic column selection and role-restricted access for admin and HOD users.	3	
2	WEEK 2	• Removed legacy standalone route handlers for project mutations, consolidating all 12 operations into the unified /api/projects PATCH endpoint with an action discriminator. • Refactored middleware.ts to eliminate redundant auth checks, consolidate rate-limiting configuration, and remove legacy redirect rules. • Added server-side payload validation and centralized error handling for all export endpoints, covering empty arrays, invalid ObjectIds, and malformed query parameters.	4	
3	WEEK 3	• Optimized the admin statistics aggregation pipeline by replacing in-memory filtering with native $match, $group, and $lookup stages. • Verified the 90-day TTL index on the Notification collection for automated purging of expired documents.	6	

		• Optimized the admin statistics aggregation pipeline by replacing in-memory filtering with native $match, $group, and $lookup stages. • Configured MongoDB connection pooling for optimal reuse across concurrent serverless API invocations and Socket.io connections.		
4	WEEK 4	• Deployed compound indexes on User (department+semester, role+department) and ProjectGroup (department+semester+academicYear, internalGuide+guideStatus), confirming COLLSCAN to IXSCAN transition via explain(). • Configured MongoDB connection pooling for optimal reuse across concurrent serverless API invocations and Socket.io connections. • Applied field-level projection on high-traffic GET endpoints to reduce network payload size.	6	
5	WEEK 5	• Implemented the guide assignment action within /api/projects PATCH — validates guide role, department match, and approval status before setting guideStatus to pending and dispatching a real-time Socket.io notification to the assigned guide. • Enforced strict RBAC and NoSQL injection sanitization on all guide assignment operations using the permissions matrix and sanitizeObject() utility. • Developed the guide accept/reject workflow: acceptance transitions project to in-progress with HOD notification; rejection clears the assignment, reverts status, and notifies the HOD with rejection remarks.	5	
Total Hours			24	

Work planned for February - 2026			
February - 2026	• Monthly/weekly report submission and grading workflow with deadline enforcement, turn-in locking, and auto-computed progress scoring. • Real-time notification system over Socket.io with user-targeted delivery and automated lifecycle event triggers. • Principal-Admin chat module with page-contextual messaging and rubric management system with department-scoped CRUD. • Admin dashboard analytics with role-adaptive data, drill-down aggregation, and per-student grade reporting. • Cloud deployment and hosting for Phase-1 internal testing and validation.		

Monthly summary :

Implemented an advanced Excel export feature with multi-criteria filtering, dynamic column selection, and integrated project/guide metadata. Streamlined the codebase by removing deprecated routes, cleaning schemas, and consolidating middleware. Enhanced database efficiency through indexing, optimized aggregations, and connection tuning. Developed the end-to-end Guide Assignment backend with real-time notifications, automated workflows, and enriched APIs, while strengthening overall security using RBAC, input validation, and NoSQL injection safeguards.

Sign of Supervisor

Remarks :

1. Successfully completed all assigned responsibilities within the stipulated timeframe.
2. Demonstrated sound understanding and consistent involvement in project activities.
3. Contributed effectively to the overall development and improvement of the system.
4. Maintained a professional, disciplined, and responsible approach throughout the reporting period.